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Coarse-grained gene expression *E. coli* model for computational "host-aware" design of synthetic biological systems

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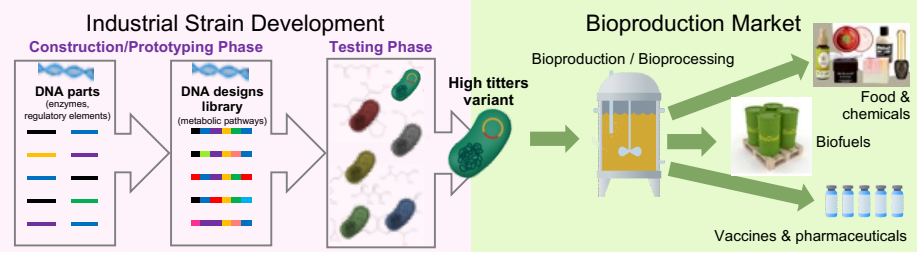
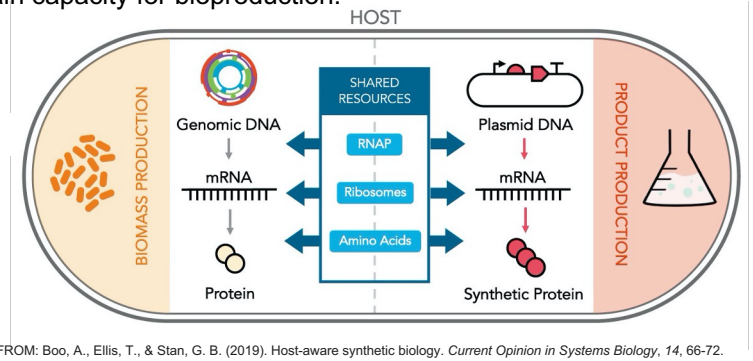
INTRODUCTION

This **biomanufacturing of products** can be achieved by genetic engineered microorganisms **(GMOs) in which transplanted synthetic DNA** encodes metabolic pathways that produce high-added value products.



Nevertheless, the transformation of microbial strains with synthetic DNA can severely **diminish cell fitness**, as the introduced system consumes a significant portion of **limited cell resources** and results in diminished strain capacity for bioproduction.



FROM: Boo, A., Ellis, T., & Stan, G. B. (2019). Host-aware synthetic biology. *Current Opinion in Systems Biology*, *14*, 66-72.

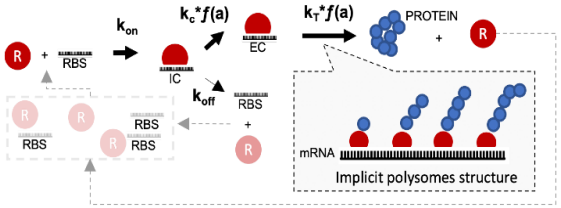
Current practises: screening methods for the construction of recombinant metabolic pathways, → impractical for large pathways. **Proposed:** forward CAD based approach → more suitable for efficient prototyping

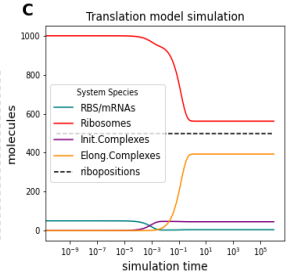
RESULTS

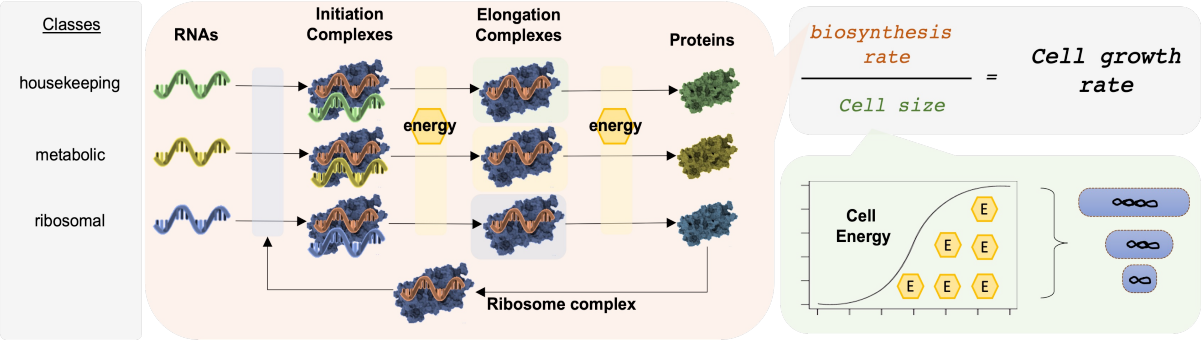
Integrated mathematical models of molecular processes

Coarse grained, gene expression cell growth :

- 13 biomolecular species + energy
- 3 high-level cell functionalities
- Limited resources: Ribosomes, Proteome, Energy
- Phenotype: Cell growth rate

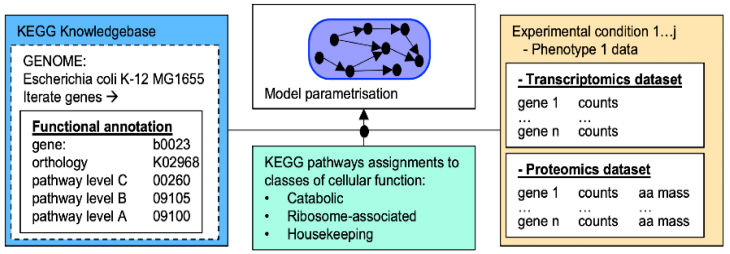
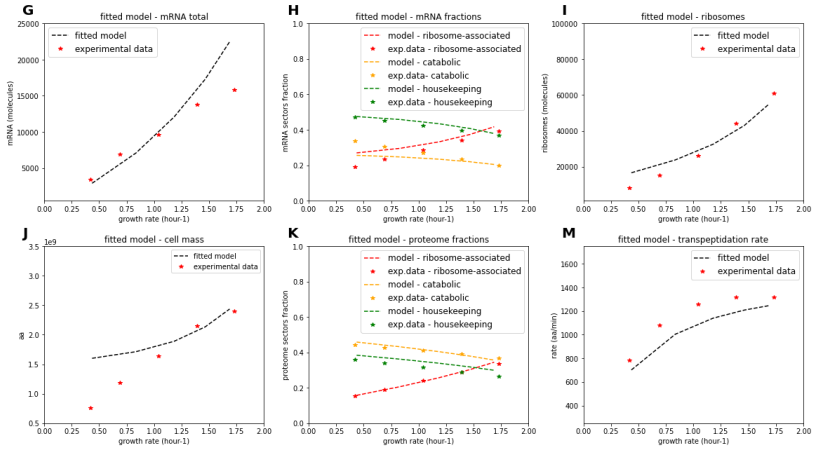






Key achievements

- Polysome structures
- Accurate/realistic ribosomes allocation
- Low computational power

Calibration of a virtual *E. coli* cell

- Model species quantification by **multi-omics datasets** & in combination with **KEGG ontology**
- Model reaction rates parametrization by **optimization algorithm** to fit to experimental data of *E.coli*'s physiology